

E3SM QuickView 2.5.1 Release (Apr. 24, 2026)

BER-ASCR SciDAC Partnership (PAESCAL), SciDAC Institute (RAPIDS)

Scientific Achievement

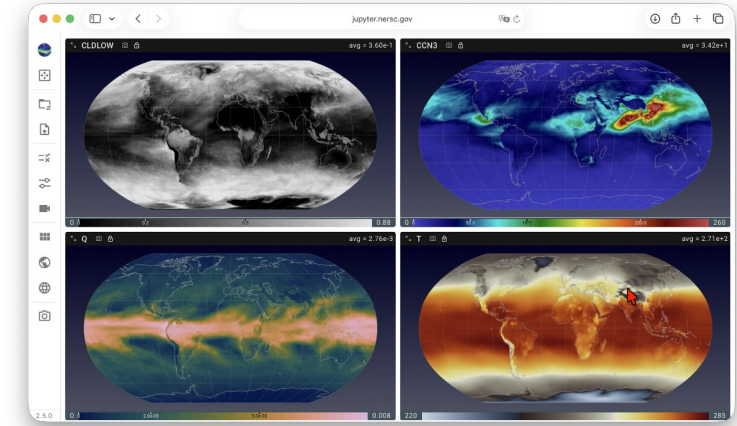
Developed a new version of the interactive visualization tool E3SM QuickView with a redesigned user interface. Deployed the tool to NERSC. Improved infrastructure layer to substantially speed up visualization of ultra-high-resolution data. Provided more general support for multidimensional arrays in the input and output files of the E3SM atmosphere and land models.

Significance and Impact

- The development and maintenance of complex numerical models like E3SM requires analysis of numerous variables and their relationships. While scripting is tedious and distracting, advanced tools are not yet widely used by Earth scientists due to steep learning curves.
- Intuitive and interactive data exploration enabled by QuickView substantially speeds up model developers' workflow for quick inspection and code debugging.
- The deep collaboration between Industry and lab – and between computer scientists and physical scientists – has developed an interface strategy for providing multiple tools for E3SM. The strategy is also generally applicable to many disciplines.

Technical Approach

- Python and Trame are used to create a simple and intuitive User Interface (UI) to ParaView, a powerful, sophisticated, and general-purpose analysis and visualization tool.
- A lightweight UI targeted at a typical workflow helps to make QuickView easy to use.



A screenshot of E3SM QuickView version 2, which offers Earth system modelers a lightweight, highly customized, and intuitive interface to ParaView. It facilitates model development without the steep learning curve of modern visual analytics.

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Publication(s) for this work:
Code repository: <https://github.com/Kitware/QuickView>
Software website: <https://kitware.github.io/QuickView/>